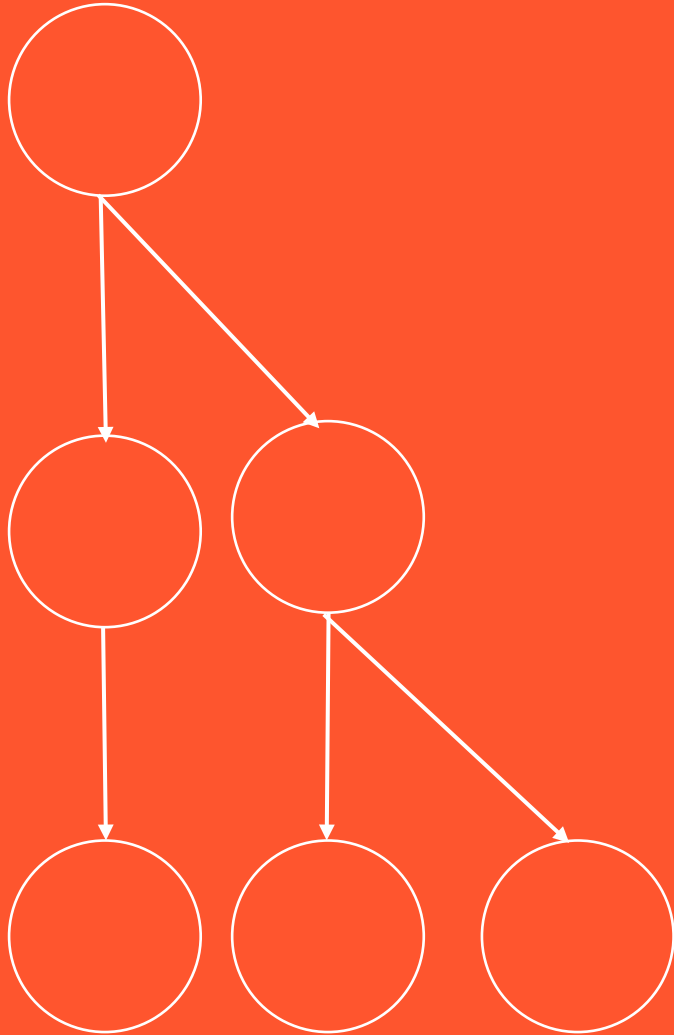


BCOG 100

Spatial Cognition

Lecture 1: Maps in the Mind



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URBANA-CHAMPAIGN

Reading Check 8-1

Password: space

Question: What's the difference between egocentric and allocentric coding, and which one do human babies have first?

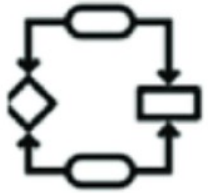


Spatial Cognition



Computation

What does the system do?



Algorithm

How does the system do it?



Implementation

How is the system realized?

Recap & Overview

Hebbian learning, Reinforcement Learning, and Pattern Recognition made vertebrates pretty smart.

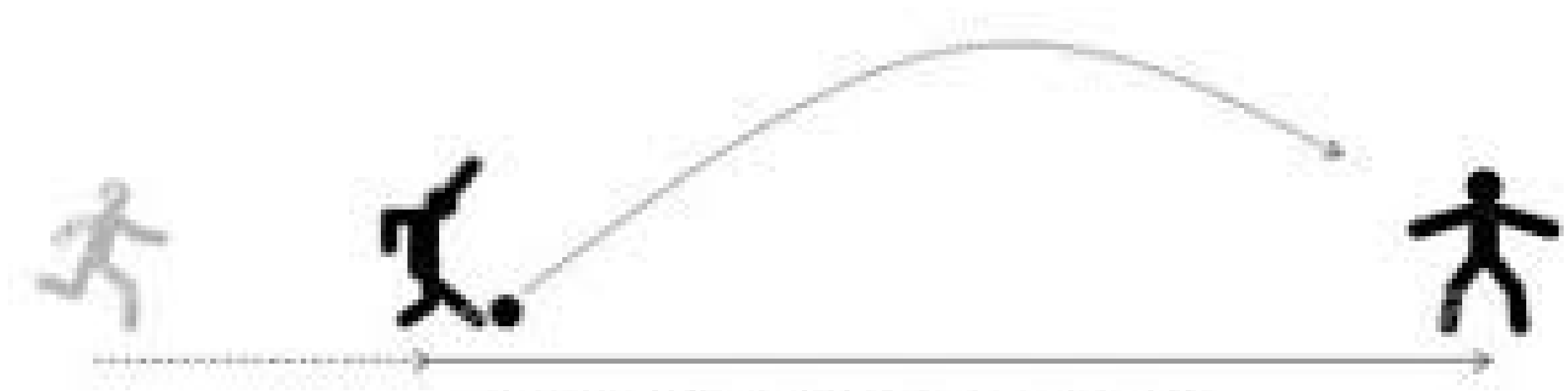
Today:

- The next challenge—stability, structure, and exploration
- Computational/Functional Level View

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Problem 1: The Problem of Self-Motion and Perceptual Stability

- When we move, the world seems to move too
- Our brains must keep perception stable amid constant change
- The same problem faced early vertebrates

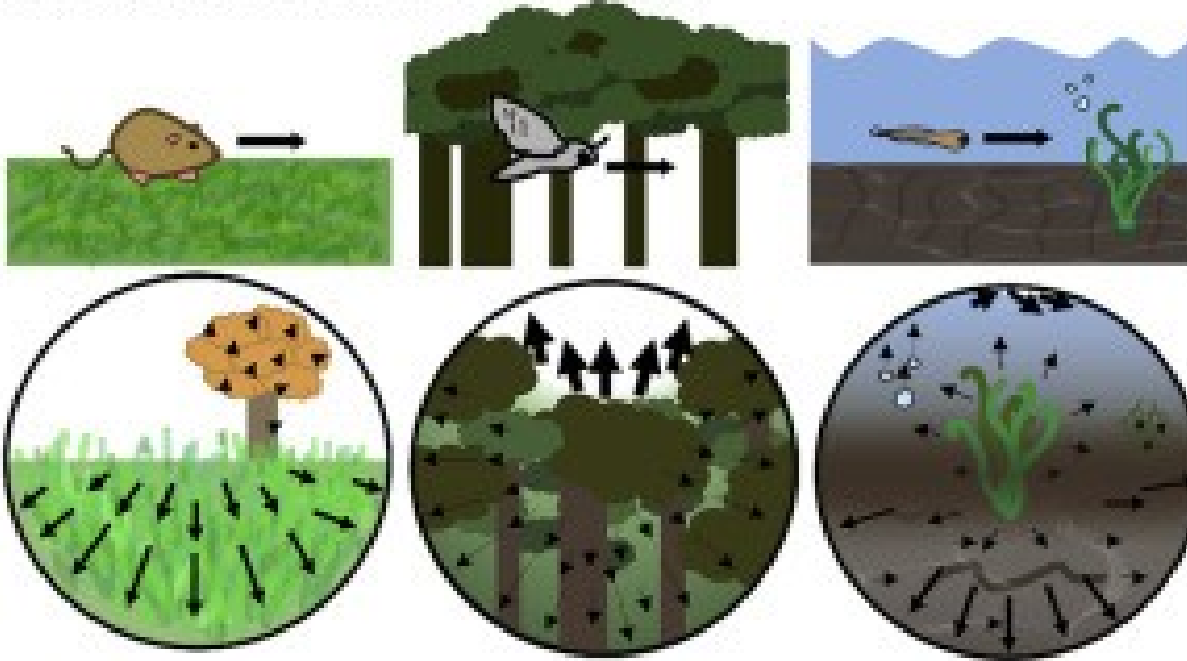


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Early Vertebrate Solutions to the Stability Problem

- Fish integrate multiple motion cues to stay oriented
- Vision, balance, and body feedback work together
- These systems evolved to keep perception stable during movement

SPATIAL PATTERNS IN OPTIC FLOW SIGNALS



ELEVATION-DEPENDENT PERFORMANCE

Spatial Cognition

Figure 1



Non-Vertebrate Comparisons

- Many invertebrates can't separate self-motion from world-motion
- Orientation depends on external cues or pauses in movement
- Vertebrates evolved internal mechanisms for stability

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Problem 1 : Summary & Key Terms

- **Challenge:** movement changes sensory input
- **Goal:** maintain perceptual stability during motion
- **Key terms:** optic flow, vestibular input, proprioception, efference copy

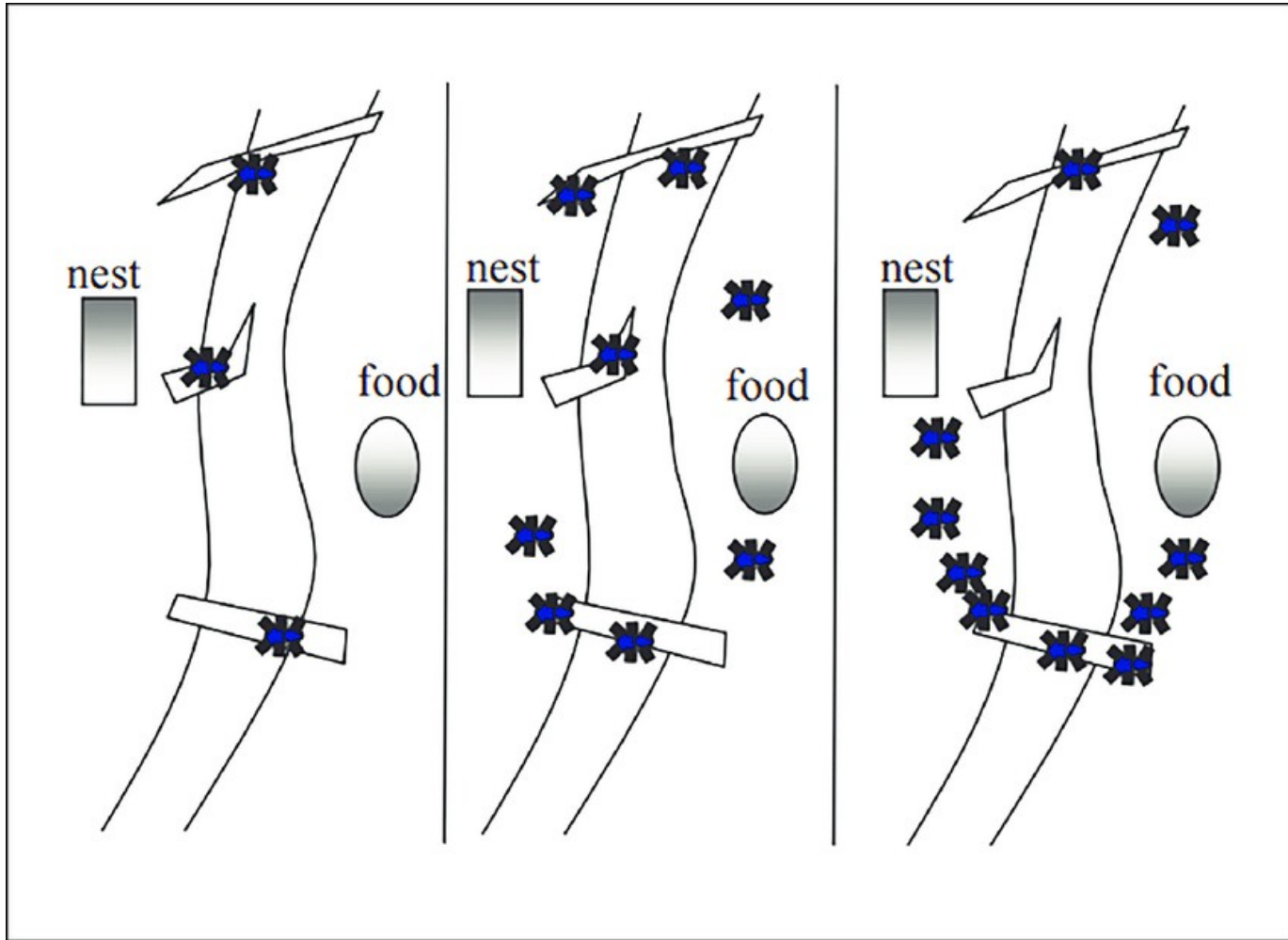
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Problem 2 : Tracking Position Without Landmarks

- How do we know where we are when we can't see?
- Navigation often depends on internal cues, not external signals
- Humans, like other animals, can estimate movement through unseen space

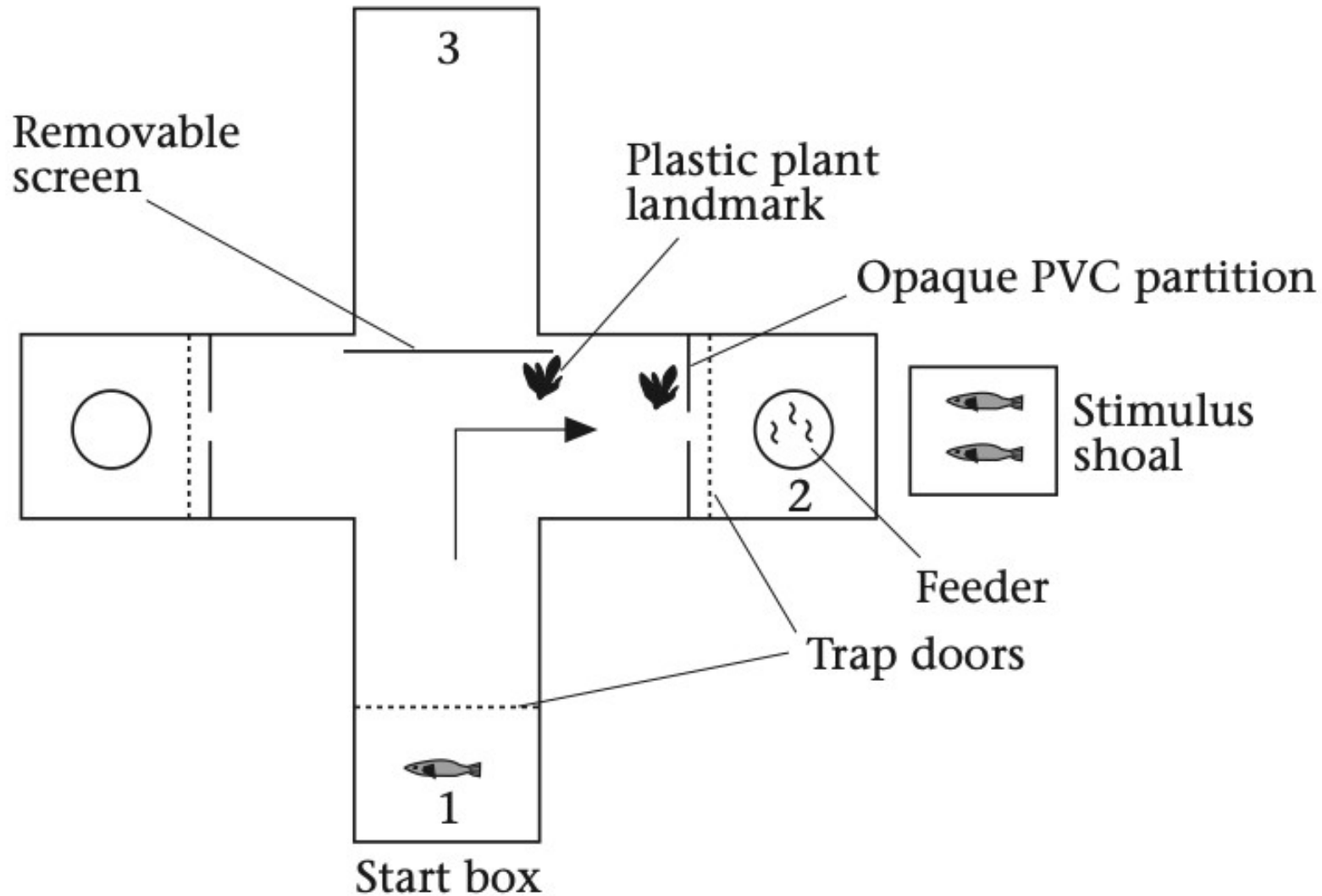
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Ant Route Learning: Sequences, not Maps

- Ants learn routes through repetition and reward
- But they often repeat old motor plans rather than using spatial relations
- Reveals limits of habitual route learning

Spatial Cognition



Fish: Learning Locations, Not Movements

- Fish learn *places*, not fixed paths
- Doing so requires three linked abilities:
- Relational mapping, perceptual invariance, and self-localization

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Problem 2: Summary & Key Terms

- **Challenge:** estimating position without visual cues
- **Two strategies:** egocentric routes vs. allocentric maps
- **Key terms:** path integration, dead reckoning, self-localization

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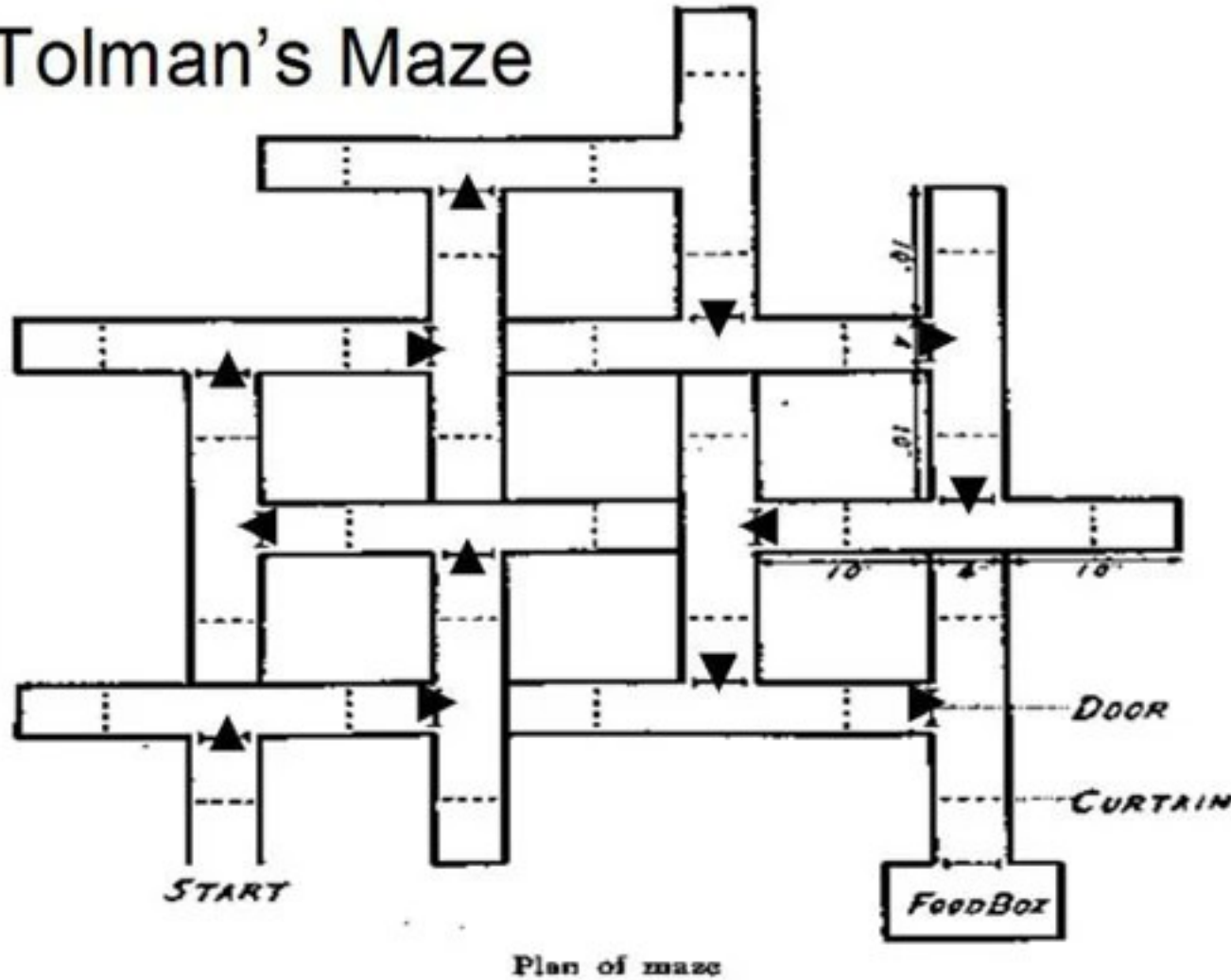


Problem 3: Latent Learning and the Hidden Map

- We can learn about space without reward or repetition
- Exploration builds knowledge that later guides behavior
- Suggests an internal “map” that exists before it’s needed

Spatial Cognition

Tolman's Maze



Tolman's Maze Experiment

- Rats explored a maze without reward, then found the goal quickly
- Learning occurred before reinforcement
- Evidence for an internal "cognitive map"

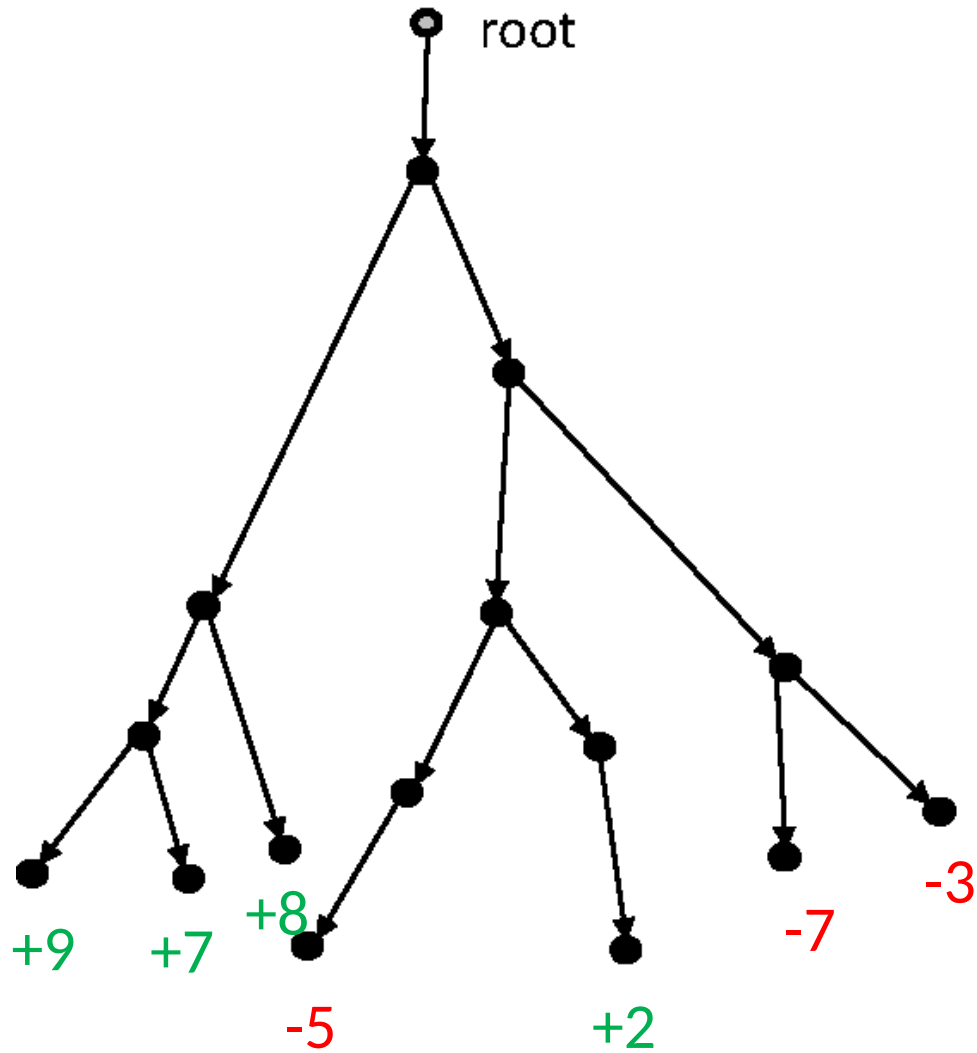
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Problem 3: Summary and Key Terms

- **Challenge:** learning about space without reward
- Latent learning reveals internal “maps” built through exploration
- **Key terms:** cognitive map, model-based learning, intrinsic motivation



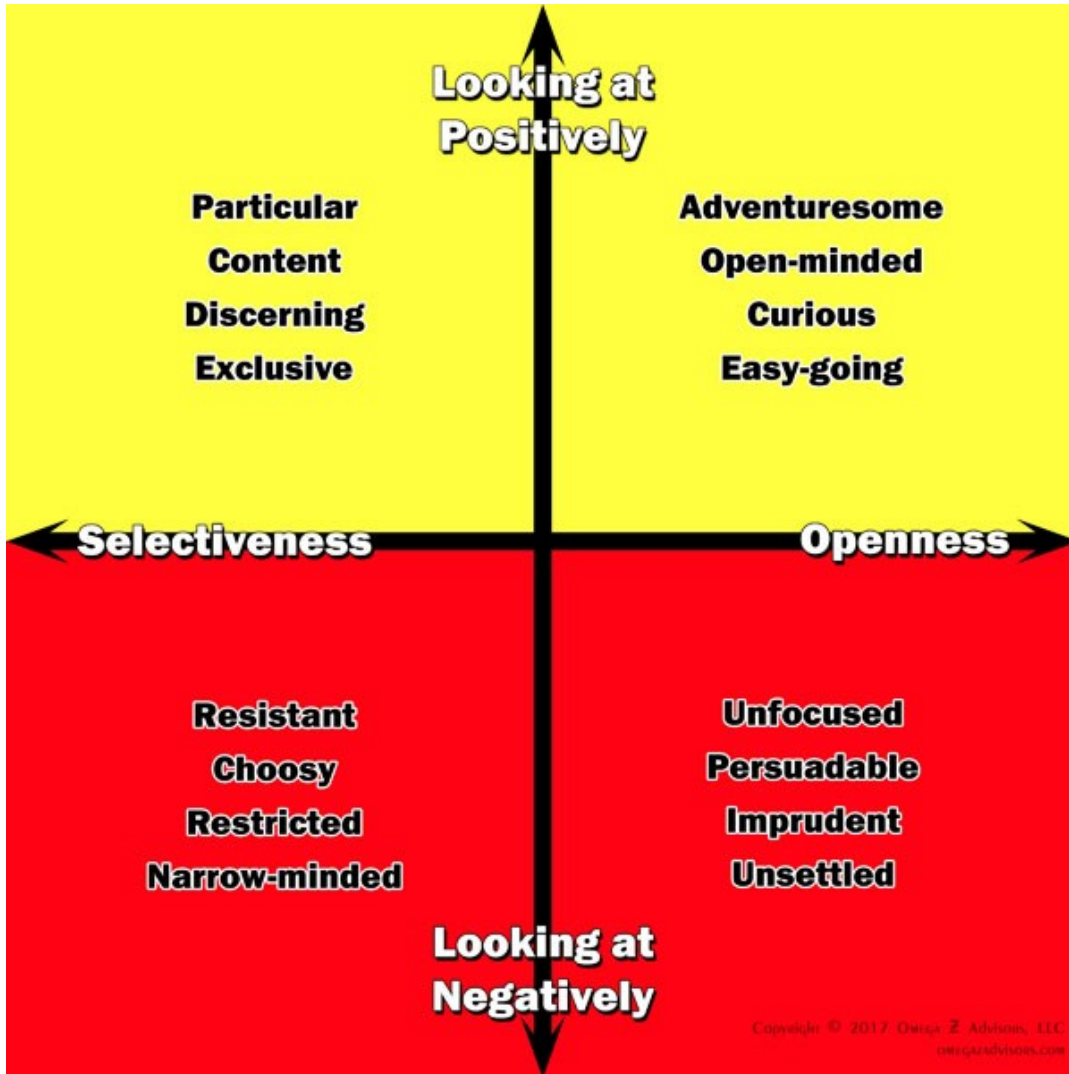
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Problem 4: The Exploration Problem

- We can stick with what we know—or try something new
- Exploration finds information; exploitation secures reward
- Curiosity helps balance the two

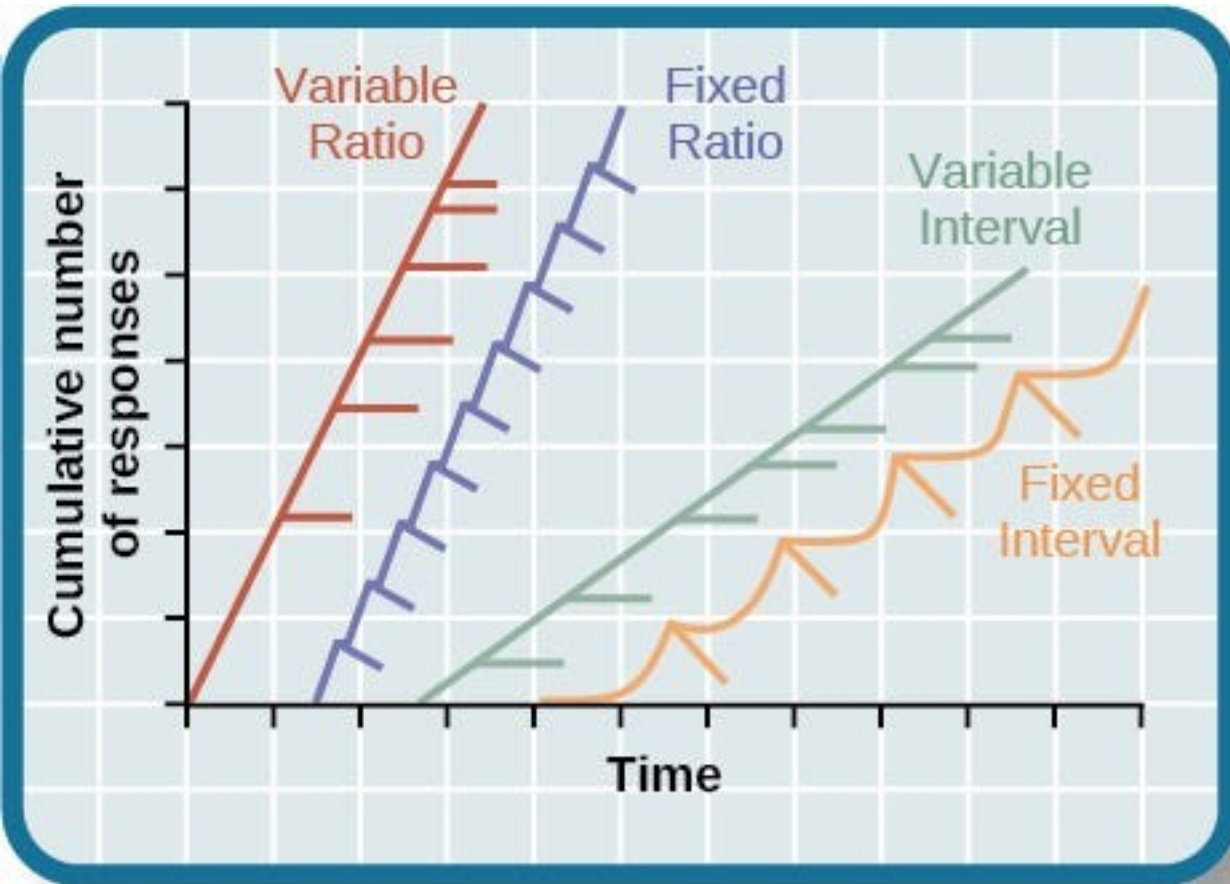
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Exploration and Exploitation in Everyday Life

- The explore-exploit tradeoff appears everywhere
- From foraging and learning to relationships and creativity
- Exploration builds models; exploitation applies them

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Gambling, Uncertainty, and Variable Rewards

- Uncertainty keeps behavior going
- Variable rewards strengthen exploration
- The same principle drives both learning and addiction

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Problem 4: Summary and Key Terms

- **Challenge:** balancing exploration and exploitation
- Curiosity motivates learning when rewards are uncertain
- **Key terms:** exploration–exploitation tradeoff, variable ratio reinforcement, prediction error



Spatial Cognition

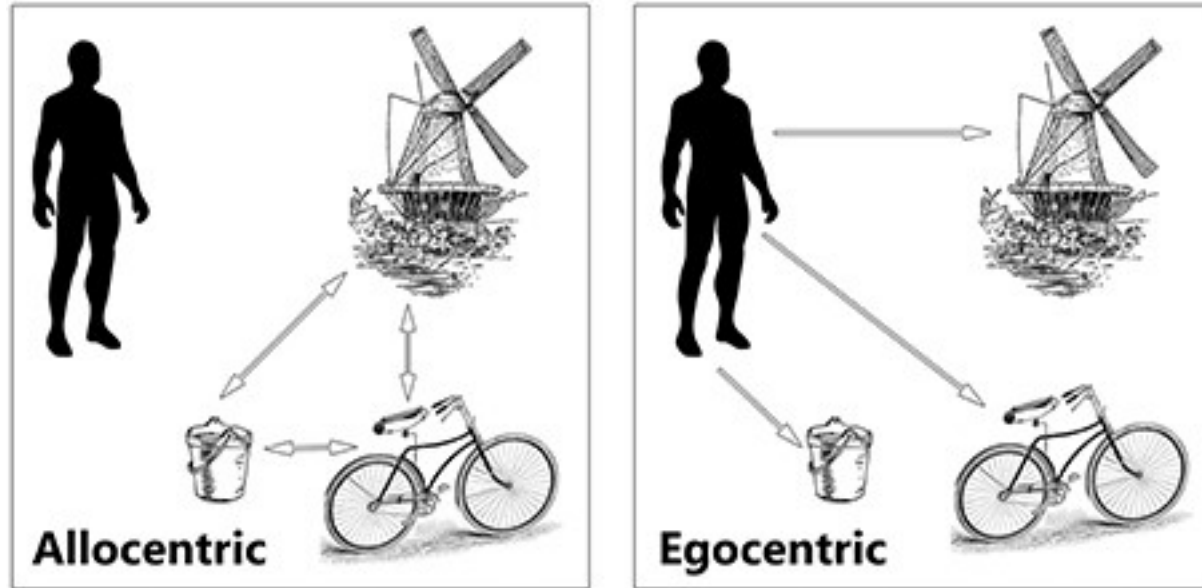


The Limits of Reinforcement and Pattern Recognition

- Reflexes, Hebbian learning, and RL enable adaptive behavior
- But they depend on past experience, not internal structure
- True navigation requires representing unseen relations

Spatial Cognition

Spatial Reference Frames



The Benefits of Spatial Mapping and Exploration

- Spatial maps enable detours, shortcuts, and flexible plans
- Curiosity expands those maps beyond immediate goals
- Together they transform learning from reactive to predictive

Spatial Cognition



Evolutionary Continuity and Ecological Payoffs

- Spatial systems are conserved across vertebrates
- Maps improve foraging, homing, and territory use
- Internal models make behavior more efficient and flexible

Spatial Cognition



Functional Summary: From Reacting to Predicting

- Early learning systems reacted to immediate outcomes
- Spatial mapping added structure and foresight
- Prediction turned behavior from reactive to proactive



Figure 1: Two ways to choose which route to take when traveling home from work on friday evening.

- Cognitive maps anticipate outcomes, not just actions
- Model-based systems use internal structure to plan
- The same principles appear in brains and algorithms

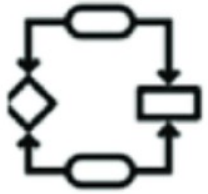
- Cognitive maps anticipate outcomes, not just actions
- Model-based systems use internal structure to plan
- The same principles appear in brains and algorithms

Spatial Cognition and Curiosity



Computation

What does the system do?



Algorithm

How does the system do it?



Implementation

How is the system realized?

Recap

- Spatial cognition and the evolution of internal models
- Curiosity as a way to fill out the model

Up Next

- Algorithms and representations for mapping, planning, and exploration